

Chapter 2

Operating diagrams

2.1 Introduction

Operating diagrams are the road maps to the operation and maintenance of an industrial or commercial power system. The diagrams can be considered as the primary tool for work on a power system. Without the guidance that is provided by the information contained on these system diagrams, the operation and maintenance of an industrial or commercial complex would be extremely difficult and potentially unsafe.

Vendor's drawings and instruction manuals, plant drawings, maintenance histories, spare parts lists, standard operating procedures, and other documents should be systematically filed and readily available to the personnel who are involved in system operation. A plant's core distribution system may have a life of 30 or more years. Complete, up-to-date equipment documentation is essential not only for supporting routine daily activity, but also for the successful implementation of system expansion or modification. System documentation should include simplified one-line drawings that show only power sources, transformers, voltage levels, major loads, disconnecting devices, switches, and breakers. These drawings are essential for the planning and illustration of operation activity. A master file should be kept in which all revisions are noted and the drawings are subsequently updated because there may be several copies or files of essential drawings that depict the plant. The use of computer-aided drafting systems may expedite the process.

The drawings and the equipment identifications should agree. Frequently, equipment identification is incomplete and poorly located. Staff reliance on such documents may result in errors, if a system is not in place to ensure accuracy. Disconnecting means, switch operators, circuit breaker control switches, circuit breakers, push buttons, motors, and other devices should have complete stand-alone identification nameplates attached in immediate full view (i.e., "FDR 12, Sub. 3" vs. "FDR 12" for a switch control handle). Identification should be unique and affixed to all associated apparatus. The location of the identification nameplate can be as critical as the accuracy of the nameplate.

Critical process operations have been accidentally shut down because circuit breakers in panelboards were not properly identified. The identification of circuits during commissioning should always be verified. A mandatory policy requiring the regular review and updating of panelboard schedules should be instituted.

No standard diagram or type of diagram that is available can suffice for all industrial plants or commercial complexes. No standard electric distribution system is adaptable to all industrial plants because two plants rarely have the same requirements. This chapter outlines the types of diagrams and information that are available to map any distribution system. Consistency is an important factor, however, in keeping diagrams readable and understandable.

2.2 Single-line diagram (one-line diagram)

A reliable single-line diagram of an industrial or commercial electrical power distribution system is an invaluable tool. It is also called a one-line diagram. The single-line diagram indicates, by single lines and standard symbols, the course and component parts of an electric circuit or system of circuits. The symbols that are commonly used in one-line diagrams are defined in IEEE Std 315-1975.¹

The single-line diagram is a road map of the distribution system that traces the flow of power into and through the system. The single-line drawing identifies the points at which power is, or can be, supplied into the system and at which power should be disconnected in order to clear, or isolate, any portion of the system.

2.2.1 Characteristics of an accurate diagram

The following characteristics should help to ensure accuracy as well as ease of interpretation:

- a) *Keep it simple.* A fundamental single-line diagram should be made up of short, straight lines and components, similar to the manner in which a block diagram is drawn. It should be relatively easy to get the overall picture of the whole electrical system. All, or as much as possible, of the system should be kept to one sheet. If the system is very large, and more than one sheet is necessary, then the break should be made at voltage levels or at distribution centers.
- b) *Maintain relative geographic relations.* In many cases, it is possible to superimpose a form of the one-line diagram onto the facility plot plan. This is very helpful toward a quick understanding of the location of the system's major components for operating purposes. It may, however, be more difficult to comprehend the overall system operation from this drawing. Such a drawing could be used for relatively simple systems. For more complex systems, however, it should be used in addition to the fundamental single-line diagram.
- c) *Maintain the approximate relative positions of components when producing the single-line diagram.* The drawing should be as simple as possible and should be laid out in the same relationship as an operator would view the equipment. The diagram does not need to show geographical relationships at the expense of simplicity.

NOTE—A site plan with equipment locations may be required to accompany the single-line diagram.

- d) *Avoid duplication.* Each symbol, figure, and letter has a definite meaning. The reader should be able to interpret each without any confusion. In this regard, equipment names should be selected before publishing the document; then, these names should be used consistently.

¹Information on references can be found in 2.4.

- e) *Show all known factors.* All details shown on the diagram are important. Some of those important details are as follows:
 - Manufacturers' type designations and ratings of apparatus;
 - Ratios of current and potential transformers and taps to be used on multi-ratio transformers;
 - Connections of power transformer windings;
 - Circuit breaker ratings in volts, amperes, and short-circuit interrupting rating;
 - Switch and fuse ratings in volts, amperes, and short-circuit interrupting rating;
 - Function of relays. Device functions used should be from IEEE Std C37.2-1991;
 - Ratings of motors, generators, and power transformers;
 - Number, size, and type of conductors;
 - Voltage, phases, frequency, and phase rotation of all incoming circuits. The type of supply system (wye or delta, grounded or ungrounded) and the available short-circuit currents should be indicated.
- f) *Future plans.* When future plans are known, they should be shown on the diagram or explained by notes.
- g) *Other considerations.* Refer to IEEE Std 141-1993 for further discussion of single-line diagrams.

2.2.2 Uses of the single-line diagram

The single-line diagram may be used in a number of important ways in operating and maintaining an industrial or commercial power distribution system. Frequently, the single-line diagram, with all of the listed information, becomes too crowded for information to be used effectively in some of the operating activities. In those instances, it is wise to produce a set of single-line diagrams, with each different diagram in the set containing the pertinent information that is required for a particular activity or set of activities. Some of the needs for special single-line diagrams are

- a) *Switching functions.* When the primary use of the diagram is to provide information to system operators for switching in order to isolate portions for maintenance or for load control, then only the data required to make the decisions necessary for system switching are included on the diagram. Sometimes, when the distribution system is complex, a separate version of the single-line diagram in block form is more usable than a complete single-line diagram. This may be identified as a "system operating diagram."
- b) *Load flow control.* This diagram is used exclusively for load flow control. It only includes the data that show system component capacities and other data that pertain to load flow.
- c) *Relaying and relay logic diagrams.* These single-line diagrams are used to describe the system protective relay systems. These diagrams are used particularly as logic and tripping diagrams that may contain a unique language used only to depict the sequence of relay or system protective component operation under various fault conditions.
- d) *Impedance diagram.* This is a single-line diagram that shows the system input impedance and the impedance of all system components in which the impedance of each circuit branch is maintained for system short-circuit analysis. This diagram should include all reactance data on large rotating apparatus or the equivalent data for groups of machines.

2.2.3 Special single-line diagrams

For a large facility that has regularly changing conditions on its electrical distribution system, a mimic bus-type, single-line diagram is very useful. This type of single-line diagram could be a fairly simple pegboard arrangement with movable circuit symbols and tapes in several colors to identify voltage levels; or it could be a light-indicating board that indicates remote switch position via telemeter information and includes load flow data. Modern systems may have this data displayed on a computer monitor.

2.2.3.1 Switching simulator

A switching simulator can be very useful for verifying the effects of any set of switching operations. Low-voltage toggle switches are substituted for circuit breakers and disconnect switches. Signal lamps indicate whether a circuit path has been omitted in the switching sequence, thereby offering the system operator a check on the switching plan prior to performing switching on the distribution system.

Interactive computer programs have been developed to provide the same system modeling as in the electrified-system one-line diagram. These programs can be rapidly manipulated to test any planned switching of an industrial plant's power system. They can be used to check a proposed set of switching instructions and, in some cases, to print a set of instructions to be used in the field.

2.2.4 Schedules

Panel schedules are not one-line diagrams by definition, but they often fulfill the same purpose. Low-voltage lighting and power panel schedules should be maintained as well as all other electrical distribution system diagrams, since they may be used in lieu of the one-line diagram.

2.3 Plan (equipment location plan)

A site plan usually is a necessary accompaniment to the single-line diagram for a complete description and mapping of the industrial and commercial electric distribution system. The locations of the major components of the system are usually easy to visualize; however, circuit routing is difficult to comprehend without a site plan.

Site plans are important for a number of reasons, all of which could affect the operation of the industrial plant or commercial complex at some time. If a major catastrophe such as fire, flood, or storm damage should occur, a site plan would be an important tool if the distribution system were to be reconstructed. The expansion and/or rearrangement of an electrical distribution system could be extremely difficult without accurate records of the location of existing system components. The site plan can be important for identifying the proximity of electrical system components to other maintenance work that may be taking place.